

5.2 Applications of Neural Networks and Deep Learning

قومی اے آئی پالیسی 2025 کے مطابق مصنوعی ذہانت پاکستان کی منصوبہ بندی کا حصہ ہے: جدت، مہارت، اخلاقی استعمال اور ڈیجیٹل ترقی۔

CS XII

Chapter 5: Application and Impacts of Computing

Lecture 07 of 15

5.2 Applications of Neural Networks and Deep Learning

LEARN IT

Machine learning, neural networks and deep learning will play a key role in Pakistan's future. The **National AI Policy 2025** highlights how artificial intelligence is now part of national planning, focusing on **innovation, skill-building, ethical use and digital growth**.

Sector	Potential impact in Pakistan
Education	Smart learning systems, online tutoring, automatic assessments, and finding where students need specific help.
Healthcare	Disease detection, medical image analysis, and managing hospital data. Crucial for rural areas lacking specialist doctors.
Agriculture	Monitoring crops, predicting weather, managing irrigation, and estimating yields — highly important for Pakistan's economy.
Business and banking	Fraud detection, customer support (chatbots), market forecasting, and intelligent risk analysis.
Local languages	Speech recognition, translation and text-to-speech tools for Urdu, Sindhi and other regional languages.

To benefit from this future, Pakistan needs skilled individuals. You should focus on **programming (such as Python)**, **mathematics**, **data analysis**, **problem-solving** and the **ethical use of technology**.

WORKED EXAMPLE

Worked example: pick a sector and write a three-line "impact paragraph" for 5 marks.

Agriculture: A cotton grower in Sanghar photographs leaves on a cheap phone. A deep-learning model trained on Pakistani crop images flags whitefly early. The farmer sprays only the affected block, saves pesticide cost, and protects yield — which matters because agriculture is a large part of Pakistan's economy.

REMEMBER IT

Five sectors: E-H-A-B-L. Education, Healthcare, Agriculture, Business/banking, Local languages. Policy words: innovation, skills, ethics, digital growth.

COMMON MISTAKE

Mistake: Writing only "AI will make Pakistan modern" with no sector or example.

Fix: Name the National AI Policy 2025, pick two sectors from the table, and give a local example (rural hospital without a specialist; Urdu/Sindhi speech tools; bank fraud).

اردو میں سمجھیں

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CHECK YOURSELF

1. Name the national policy mentioned for AI in Pakistan.
2. Why is AI in healthcare especially important for rural areas?
3. List three skills students should learn to prepare for this future.

Answers: 1. National AI Policy 2025. 2. Those areas often lack specialist doctors; AI can help with disease detection and medical image analysis. 3. Any three of: programming (Python), mathematics, data analysis, problem-solving, ethical use of technology.

LEARN IT

Neural networks and deep learning are widely used in modern life. Many intelligent systems use them in the background to **recognise information, make predictions and support users**.

#	Application	What it does	Example
1	Image recognition	Identifies objects, faces or patterns in images. Used in security cameras, photo search and biometric locks.	Unlocking a smartphone using face recognition; a camera at a Karachi plaza matching a face with a staff list.
2	Speech recognition	Converts spoken words into digital text or commands. Deep learning helps with accents, pronunciation and voice patterns.	A student speaks to Google Assistant or Alexa to set a morning alarm; Urdu voice search on a phone.
3	Natural Language Processing (NLP)	Helps computers understand, process and respond to human language grammatically and in context. Core of chatbots, translators and AI writers.	Using Google Translate to turn a difficult English paragraph into Urdu; a bank chatbot answering in Urdu.

#	Application	What it does	Example
4	Healthcare systems	Analyse complex medical images (X-rays, CT scans, MRIs) to detect diseases early and accurately.	A hospital system highlighting abnormal patterns in a lung X-ray that might indicate pneumonia or tumours — useful at a DHQ hospital that has few radiologists.
5	Recommendation systems	Analyse past behaviour and preferences to suggest content, products or services.	A student watches computer-science videos on YouTube; the platform recommends more programming tutorials.
6	Autonomous vehicles	Self-driving cars use neural networks to process real-time camera and radar data: roads, signals, pedestrians, obstacles.	A self-driving car detects a pedestrian crossing and automatically applies the brakes.
7	Fraud detection	Banks compare current activity with previous spending patterns to catch unusual transactions.	An HBL or Meezan app flags a sudden Rs 80,000 payment from a new city at 3 a.m.
8	Assistive technologies	AI supports people with disabilities.	Speech-to-text tools for users who have difficulty typing; screen readers that speak Urdu text aloud.

WORKED EXAMPLE

Worked example: match the app to the application name — a day in Ahmed's life in Hyderabad.

1. 7:00 am — phone unlocks by face. → **Image recognition**.
2. 7:05 am — "Ok Google, set alarm for Maghrib." → **Speech recognition**.
3. 8:00 am — Google Translate helps with an English article. → **NLP**.
4. 4:00 pm — YouTube suggests a Python tutorial after he watched Chapter 5 notes. → **Recommendation system**.
5. 9:00 pm — JazzCash texts "We blocked a payment that does not match your pattern." → **Fraud detection**.

REMEMBER IT

Eight applications: I-S-N-H-R-A-F-A. Image, Speech, NLP, Healthcare, Recommendations, Autonomous vehicles, Fraud, Assistive tech. "I See New Hospitals Recommend Autonomous Fraud-Assist."

COMMON MISTAKE

Mistake: Mixing up speech recognition and NLP.

Fix: Speech recognition turns **sound into text** (or a command). NLP understands the **meaning of language** (grammar, context, translation, chat replies). You can have NLP on typed text with no microphone at all.

نیورل نیٹ ورکس اور ڈیپ لرننگ کی مشہور ایپلی کیشنز یہ ہیں: تصویر کی پہچان، آواز کی پہچان، قدرتی زبان کی پروسیسنگ (NLP)، صحت کے نظام، تجویز کرنے والے نظام، خود چلنے والی گاڑیاں، فراڈ کی نشاندہی اور معاون ٹیکنالوجیز۔ روزمرہ میں چہرے سے فون کھولنا، یوٹیوب کی تجاویز اور بینک کا مشکوک لین دین پکڑنا سب اسی خاندان سے ہیں۔

CHECK YOURSELF

1. Which application converts spoken words into text or commands?
2. Give the source example of a recommendation system.
3. How do self-driving cars use neural networks?

Answers: 1. Speech recognition. 2. YouTube recommending more programming tutorials after a student watches computer-science videos. 3. They process real-time camera and radar data to understand roads, traffic signals, pedestrians and obstacles (for example braking for a pedestrian).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Vision applications include:

- a. Face unlock, medical images, number-plate reads
- b. Primary keys
- c. Bubble sort
- d. OSI ping

MCQ 2. Language applications include:

- a. Translation, speech-to-text, chatbots
- b. NAND grouping
- c. Foreign keys
- d. Pie charts of cables

MCQ 3. Healthcare DL must still:

- a. Be checked by clinicians
- b. Replace doctors unsupervised always
- c. Ignore bias
- d. Train on one patient

MCQ 4. A local example could be:

- a. Urdu voice assistants / crop images
- b. K-maps of farms
- c. Access reports of OSI
- d. Figma of NAND

MCQ 5. Recommendation systems are:

- a. ML/NN apps (what to watch/buy)
- b. Queues in printers only
- c. Stacks of plates only
- d. ER only

MCQ 6. The exam wants:

- a. Named field + how NN/DL helps + one risk
- b. Only the word Pakistan
- c. Only careers list
- d. Only math of backprop

Extra short questions

1. Four application areas from the book.
2. One risk in health.
3. One risk in security/surveillance.
4. Why "application" questions still need a definition line.

Extra long question

1. Pick vision, health, language. For each: system, data it needs, a benefit, a harm. 10-mark table.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Vision, speech, health, transport, security, education — any four with a concrete system.
2. False negative on an X-ray; bias against a skin tone.
3. Misidentification; privacy.
4. 1 mark for the term, rest for use+limit.

Long

1. Face gate / images / fast entry / racism in scores. X-ray / labelled scans / earlier catch / missed cases. Urdu STT / audio / inclusion / leaking class recordings.